

J A V A S C R I P T

Unit 1: Fundamentals

Variables · Conditionals · Loops · Arrays · Functions

The most popular and widely-used programming language in the world

 Client-side |  Server-side (Node.js) |  Mobile Apps



What is JavaScript?

Dynamic Language

Interpreted at runtime — no compilation needed. Runs directly in the browser.

Web Standard

One of the three core web technologies: HTML (structure), CSS (style), JS (behavior).

Versatile

Used for front-end (DOM manipulation), back-end (Node.js), mobile apps, and games.

Event-Driven

Responds to user actions: clicks, key presses, form submissions, timers.

Variables & Data Types

Keyword	Scope	Re-assign	Re-declare	Best Use
var	Function	✓ Yes	✓ Yes	Legacy code
let	Block	✓ Yes	✗ No	Variables that change
const	Block	✗ No	✗ No	Constants & objects

```
// var - function  
scoped (avoid in  
modern JS)  
var name =  
"JavaScript";
```

```
// let - block  
scoped,  
reassignable  
let version = 1;  
version = 2; //  
OK
```

```
// const - block  
scoped, cannot  
reassign  
const PI =  
3.14159;  
// PI = 3; // ✗  
TypeError
```

```
// Data types  
let str = "Hello";  
// string
```

Operators in JavaScript

Arithmetic

- + Addition
- - Subtraction
- * Multiplication
- / Division
- % Modulo
- ** Exponent

Comparison

- == Equal (loose)
- === Equal (strict)
- != Not equal
- !== Strict not equal
- > Greater than
- < Less than

Logical

- && AND
- || OR
- ! NOT

Assignment

- = Assign
- += Add & assign
- -= Subtract & assign
- *= Multiply & assign
- /= Divide & assign

Conditional Statements

if Executes block if condition is true

if...else Alternative block if condition is false

else if Chain multiple conditions

switch Match value against multiple cases

ternary ?: Compact one-line conditional

```
let day = new Date().getDay();

// if / else if / else
if (day === 0) {
  console.log("Sunday - Rest day!");
} else if (day === 6) {
  console.log("Saturday - Weekend!");
} else {
  console.log("Weekday - Work hard!");
}

// switch statement
switch(day) {
  case 5: console.log("Friday!"); break;
  case 6: console.log("Saturday!"); break;
  default: console.log("Other day");
}

// ternary operator
let msg = day === 5 ? "TGIF! 🎉" : "Keep going!";
```

Loops & Iteration

while

Loops while condition is true. Check BEFORE first iteration.

do...while

Executes block ONCE first, then checks condition.

for

Classic counter loop — init, condition, increment.

for...in

Iterates over object properties (keys).

for...of

Iterates over iterable values (arrays, strings, sets).

```
// for loop
for (let i = 1; i <= 5; i++) {
  console.log("Number: " + i);
}
```

Arrays

push()

Add to end

pop()

Remove from end

shift()

Remove from start

unshift()

Add to start

splice()

Add/remove at index

slice()

Copy portion

indexOf()

Find index of value

includes()

Check if value exists

join()

Combine to string

reverse()

Reverse in place

sort()

Sort elements

length

Number of elements

```
let colors = ["Red","Green","Blue"];
colors.push("Yellow"); // ["Red","Green","Blue","Yellow"]
colors.pop();          // removes "Yellow"
console.log(colors.length); // 3
```

Functions

Function Declaration

Hoisted — can be called before definition

Arrow Function (ES6)

Compact syntax, no own "this" binding

Function Expression

Not hoisted — must define before calling

Default Parameters

Provide fallback values for arguments

```
// Declaration (hoisted)
function greet(name) {
  return "Hello, " + name + "!";
}

// Expression
const square = function(n) { return n * n; };

// Arrow (ES6)
const add = (a, b) => a + b;
const double = n => n * 2; // one param, no parens

// Default parameters
function welcome(name = "Guest") {
```

Objects

Properties

Key-value pairs that store data

this

Refers to the current object context

Bracket notation

obj["property"] — dynamic key access

Methods

Functions stored as object properties

Dot notation

obj.property — simple access

Object.keys()

Returns array of property names

```
const person = {
  name: "Clark Kent",
  age: 36,
  greet: function() {
    return "Hi, I'm " + this.name;
  }
};
console.log(person.name);    // Clark Kent
console.log(person["age"]);  // 36
console.log(person.greet()); // Hi, I'm Clark Kent
```

Unit 1 — Quick Reference Cheat Sheet

Variables

- var — function scope (legacy)
- let — block scope, mutable
- const — block scope, immutable

Conditionals

- if / else if / else
- switch / case / default
- ternary: cond ? a : b

Loops

- for, while, do...while
- for...in (object keys)
- for...of (iterable values)

Arrays

- push/pop, shift/unshift
- slice, splice, indexOf
- length, sort, reverse

Functions

- declaration (hoisted)
- expression / arrow (=>)
- default & rest params

Objects

- key:value properties
- methods with this
- dot & bracket notation